

Kaustub P Anand

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RESEARCHER BIO

I am a PhD candidate at Purdue University in the Department of Physics and Astronomy, and working in the Department of Earth, Atmospheric, and Planetary Sciences (EAPS). My research interests are in solar system dynamics, especially planetary rings, moons, comets, and small bodies (ex: Centaurs). I primarily work with numerical simulations and have done observations in the past.

TECHNICAL SKILLS

Coding : Python, FORTRAN, NetCDF, HDF5, SHTools, Java, IDL, $\text{\LaTeX}$, SAOImage DS9, C, C++, Matlab, IRAF

Physical Effects Coded: Gravitational harmonics, collisional fragmentation, radiation forces including Yarkovsky and the Yarkovsky-Schach effect, 1D fluid debris rings, hybrid fluid ring – N body code

EDUCATION

Purdue University Aug 2021 – Present
Doctor of Philosophy (PhD) in Physics.
Thesis: Understanding the origins and dynamics of rings and corresponding satellites with application to Centaurs and the Martian moons.
Advisor: Dr. David A. Minton

Purdue University Aug 2017 – May 2021
Bachelor of Science in Physics Honors.
Minors in Astronomy and Math; Graduated with Distinction.
Cum. GPA: 3.92/4.00; Major GPA: 3.96/4.00

PUBLISHED RESEARCH PAPERS

Ejected Surface Regolith as a Potential Source Material for Centaur Rings doi:10.3847/PSJ/ae7724
Kaustub P. Anand, David A. Minton, Julie Brisset

The Sesquinary Catastrophe on Deimos Can Reconcile Its Excited Past with Its Dynamically Cool Present doi:10.3847/PSJ/ae2fba
Kaustub P. Anand, Matija Ćuk, David A. Minton

Two Possible Orbital Histories of Phobos doi:10.3847/PSJ/adclba
Matija Ćuk, Kaustub P. Anand, David A. Minton

Galaxy populations in protoclusters at cosmic noon doi:10.1051/0004-6361/202452628
Moirá Andrews, M. Celeste Artale, Ankit Kumar, Kyoung-Soo Lee, Tess Florek, Kaustub P. Anand, et al.

Swiftest: An N-body Integrator for Gravitational Systems doi:10.21105/joss.05409
Carlisle Wishard, Jennifer Pouplin, Jacob Elliott, Dana Singh, Kaustub P. Anand, David A. Minton

OTHER RELEVANT RESEARCH EXPERIENCE

Evolving a fluid Martian ring under the Yarkovsky-Schach effect Feb 2026 – Present
Advisors: Dr. David A. Minton and Dr. Matija Ćuk
Purdue University

Star formation in dust-obscured Galaxy Clusters in far-IR & protoclusters in simulation May 2019 – May 2022
Advisors: Dr. Kyoung-Soo Lee and Dr. Maria Celeste Artale
Purdue University

CONFERENCES AND INVITED TALKS

Colloquium Seminar at Southwest Research Institute (SwRI) Nov 2026

AAS Division on Dynamical Astronomy (DDA) 2024 - 2026

AAS Division of Planetary Sciences (DPS) 2024, 2026

Purdue EAPS Planetary Science Seminar (1 talk/year) 2023 - 2026

Gordon Research Seminar & Conference: Origins of the Solar System June 2025

NOIRLab: The Solar System in Context Sept - Oct 2025

Division of Planetary Sciences-Europlanet Science Congress 2023, 2025

Astronomy Seminar at the University of Connecticut Feb 2025

Outreach Talk at Nahar International School, Mumbai, India Dec 2025

Purdue University Department of Physics and Astronomy, and EAPS 1-minute colloquium

2022 - 2026

Purdue University Department of Physics and Astronomy annual poster symposium

2023

Purdue Fall Research Expo

Fall 2020

LEADERSHIP AND PROFESSIONAL EXPERIENCE

Local Organizing Committee Member for DDA 2027

Sept 2026 - Present

LOC member

Serving as a member of the Local Organizing Committee for the Division on Dynamical Astronomy (DDA) 2027 meeting in Bloomington, IN

JPL Planetary Science Summer School

May - July 2026

Proposal Manager (Management Hex Role)

Jet Propulsion Laboratory

Designed a New-Frontiers class NASA mission on Ceres called Proserpina in 11 weeks as a part of the JPL Mission Design Schools. I helped with personnel management, moderated conflict resolutions, managed graphics deliverables, and maintained flexibility to help the PI resolve last-minute problems.

Graduate Student Ombudsperson

May 2026 - Present

Ombudsperson

Purdue University

Served as the graduate student ombudsperson for the Department of Earth, Atmospheric, and Planetary Science to listen and help student grievances and conflicts.

NASA Review Panel Executive Secretary

2026

Executive Secretary

Online

Served as the executive secretary on 2 NASA proposal review panels for the NASA Science Mission Directorate.

EAPS Planetary Science Seminar Organizer

May 2025 - 2026

Crater Café organizer

Purdue University

Managed and organized the schedule and speakers of Crater Café, the weekly Purdue EAPS planetary science seminar.

Physics & Astronomy Teaching Assistant

Aug 2018 - Mar 2020 & Aug 2021 - Dec 2022

Teaching Assistant

Purdue University

Led and aided students in labs and recitations for modern mechanics (PHYS 172), electricity and magnetism (PHYS 272), senior level Computational Physics Lab (PHYS 580), and Optics Lab (PHYS 450). Started as a Graduate Teaching Assistant in Aug 2021.

Saturday Morning Astrophysics (SMAP)/Physics Inside Out (PIO)

Jan - Mar 2020; Jan 2021 - May 2022; June 2025 - 2026

Outreach Teacher

Purdue University

Teach and introduce common astronomy and physics concepts to elementary & middle school students through hands-on experiments monthly on Saturdays (SMAP) and in the June annually (PIO).

Peer Success Coach

Jan 2021 - May 2021

Tutor

Purdue University

Tutor and coach students at the Purdue Academic Success Center to improve their studying behaviour and motivate them to work towards achieving their current and new academic goals.

International Student Ambassador

May 2020 - May 2021

Ambassador

Purdue University

Aid the Purdue International Students and Scholars (ISS) office as a student ambassador in organizing events for the international student community on campus including new student orientations, cultural mixers, Q&A sessions, etc.

Global Science Partners Leadership

Aug 2019 - May 2021

Team Leader

Purdue University

Organized and led events to analyze the effects of inter-cultural interactions in professional and scientific collaborations among the members of Global Science Partners learning community (~160 students)

AWARDS & CERTIFICATES

- Purdue Graduate Student Government Travel Grant (2025)
- Purdue EAPS Departmental Travel Grant (2025)
- Hartmann Student Travel Grant for DPS (2024)
- Martin C. Jishcke Outstanding International Student Award (2021)
- Learning Beyond the Classroom Certificate for intercultural learning in Science (2021)
- Dean's List and Semester Honors (2017 - 2021)
- Honors College (2018 - 2021)
- Arthur N. Pozner Memorial Scholarship for academic excellence in Physics (2020)